

1. Circle Y has points X , M , and R on the circle with $m\angle MYX = 37.2^\circ$. If $m\widehat{MR} = 146.8^\circ$ and $\widehat{MR}$ does not intersect $\widehat{MX}$, what is $m\angle RYX$?

$$m\angle RYX = 176^\circ$$

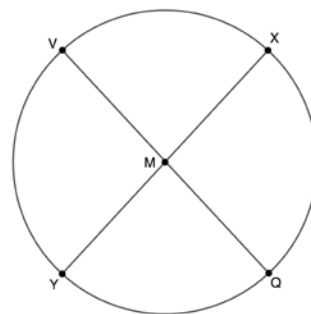
2. Circle G has radii $\overline{GH}$ and $\overline{GK}$. Find the value of x if $m\angle KGH = 122^\circ$ and $m\widehat{KH} = (5x + 7)^\circ$.

$$5x + 7 = 122$$

$$5x = 115$$

$$x = 23$$

For questions 3-4, Circle M has diameters $\overline{VQ}$ and $\overline{XY}$ that both intersect at point M .



3. Given $m\widehat{QY} = 80^\circ$, find the $m\widehat{VX}$.

$$m\widehat{VX} = 80^\circ$$

4. Given $m\widehat{QX} = 100^\circ$, find the $m\widehat{YV}$.

$$m\widehat{YV} = 100^\circ$$

For questions 5-6, consider the following context.

Lorenzo is creating a folding fan. He needs to determine the measure of the arc of the fan before he can cut the paper. He uses the expression $(5x + 119)^\circ$ as the measure of the central angle, $\angle JQZ$. For the $m\widehat{ZJ}$, he uses the expression $(25x + 59)^\circ$.

5. What is the value of x ?

$$5x + 119 = 25x + 59$$

$$60 = 20x$$

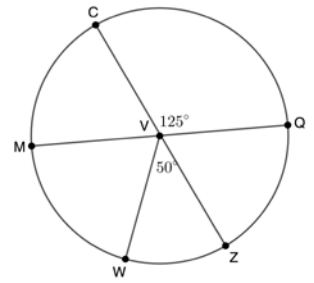
$$3 = x$$

6. What is the degree measure of the arc of the fan, $\widehat{ZJ}$?

$$134^\circ$$

7. In Circle V , $m\angle CVQ = 125^\circ$ and $m\angle ZVW = 50^\circ$. Calculate the $m\widehat{CQW}$, given that V is the center and $\overline{CZ}$ and $\overline{MQ}$ are diameters.

$$m\widehat{CQW} = 125 + 55 + 50 = 230^\circ$$

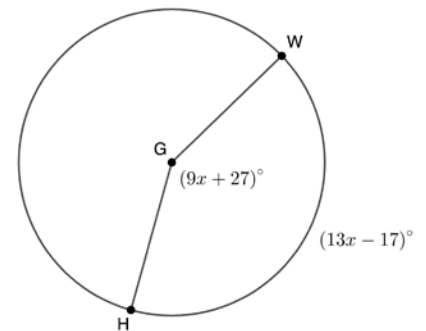


8. Circle Y has points X , Q , and W on the circle. The measure of the central angle, $\angle WYQ = 65^\circ$, $m\widehat{XW} = 117^\circ$, and $\widehat{XW}$ does not intersect $\widehat{WQ}$. Find $m\angle XYQ$.

$$360 - 65 - 117 = 178^\circ$$

For questions 9-10, consider the following context.

A horse is tied to a post at point G and is grazing in a circular pattern. She decides to walk from her position at point H to the trough of water at point W along the circumference of the circular area she can roam. The measure of the central angle, $\angle WGH$, is $(9x + 27)^\circ$, and the measure of the arc that the horse walked, $\widehat{WH}$, is $(13x - 17)^\circ$.



9. Find the value of x .

$$9x + 27 = 13x - 17$$

$$44 = 4x$$

$$11 = x$$

10. Find the measure of the arc that the horse walked, $\widehat{WH}$.

$$13(11) - 17 = 126^\circ$$